DSA ki Baithak

CODE BAITHAK

BY NAMAN BHAIYA

Recap (Previous Lecture Connection):

* Programming Language
* High Level Language vs Low Level Language
* Compiler vs Interpreter
* Java Architecture
* JDK, JRE, JVM
* Java Setup
* First Program

Java is an Object-Oriented Programming Language.

"public class Main"

* public - Access Modifier (means ye class har jagah accessible hai)

* class - Class is a user-defined data type (like a template/ blueprint)

* Main - Class name (File name same hona chahiye)

"public static void main()"

* main() - Entry point of program. It is a method.
  (Java program starts from main() only)

* static - It means method belongs to class, not object.
  (Taaki JVM bina object banaye program Start kar sake)

* void - It is the return type of main().
  (Void means nothing will be returned)

Output in Java

① System.out.print()

- System → Predefined Class
- out → Static object
- print() → Method

② System.out.println()

System.out
↓ (default)
outputs in Command line

print → cursor in same line

println → cursor in next line

Taking Input in Java

```
Scanner sc = new Scanner(System.in)
```

Predefined class

System.in
↓ (default)
takes input from Keyboard

```
int num = sc.nextInt()
String text = sc.next()
```

Variables in Java

Variable is a container that stores data. (It is not a value)

or

Variable is a named reference to a memory location that stores a value.

① Primitive Types Variable - stores directly the value
(int, double, char, boolean)

② Reference Types Variable - stores the address of value
(Non-Primitive) (String, Arrays, Objects)

String class → toString() overrided

Java Variable Naming Rules (Compiler Rules)

* Allowed characters are - Letters, Digits, Underscore, Dollar Sign
  (a-z, A-Z) (0-9) (_) ($)

* Cannot start with a digit.
  1age ✗ age1 ✓

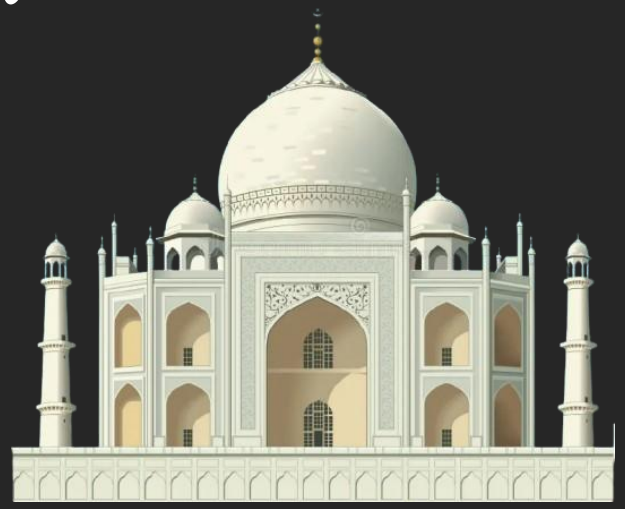
* Space is not allowed.
  my age ✗ myage ✓

* No other special character
  (@, #, !, %, &, ...) ✗

* Cannot be a Java Keyword
  class ✗
  Class ✓

(They are case-sensitive)

Keywords in Java



* Keywords are the reserved words in Java.
* Cannot be used as variable, class & Function name.
* Total approx 50+ Keywords in Java.

(public, static, class, double, return
if, else, int, void, ----)